SriUjjwal Reddy Beereddy

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Education

Arizona State University (4.0 GPA) Tempe, AZ
Bachelor of Science in Computer Science *Expected Graduation: May 2026*

Experience

AI/ML Software Engineering Intern Jul 2024 – Present
Geometric Media Lab *Tempe, AZ*

- Boosted **gunshot detection** project's accuracy **from 20% to 80%** by adapting existing machine learning models and deploying the model on a **Raspberry Pi** unit for field use in the South American rainforest, supporting **anti-poaching efforts**.
- Built ML pipelines and architectures for experimentation, using **pretrained VLM models** like **Gemini VLM API**, **Hugging Face**, and stable video diffusion models with **computer vision** across 6+ research projects for rapid prototyping and deployment of ML solutions.
- Developed **ML models** and efficient **pipelines** using **Python**, **PyTorch**, and **CUDA**; optimized large-scale model training/inference on the **SOL GPU cluster** with multiple **A100 GPUs**, achieving significantly faster inference through GPU acceleration and utilizing **wandb** for experiment tracking and seamless production rollouts.

Software Engineer Aug 2024 – May 2025
Mesa Historical Museum (EPICS) *Tempe, AZ*

- Built an interactive 3D museum exhibit using **React**, **Three.js**, and Vercel, digitizing and showcasing detailed stories of historic individuals and landmark places as immersive web experiences for enhanced visitor engagement.
- Led a **7-member team** to digitize museum archives, enabling visitors to explore collections online and increasing accessibility through agile practices and direct client communication.
- Optimized for **cost** and **performance** by developing and deploying a static-site solution with no ongoing maintenance, using JSON data and **Vercel** for fast, reliable access in the museum.

Software Engineering Intern Jan 2023 – Aug 2024
ASU Biodesign Institute *Tempe, AZ*

- Developed a **Django/FastAPI backend** to automate DNA-PAINT image analysis for 100+ datasets, enabling researchers to process 2000+ nanorobot entities per image with **ML-driven clustering** and tracking.
- Reduced analysis time from **4 hours to 10–30 minutes** per image by building a scalable pipeline and REST API, delivering a user-friendly web interface for machine learning analysis and visualization.
- Collaborated with **engineers and researchers** to maintain and document the codebase, support onboarding, and deliver analytics scripts for impactful insights.

Projects

Puter OS | *Open-Source Contribution to 33.7K+ star project* | ∼40 hours | github.com/HeyPuter/puter

- Implemented the first real-time **notification system** for Puter OS, an **open-source** browser-based operating system, using Node.js, Express, SQL, JavaScript, and Docker.
- Developed a persistent notification system using database storage, where previously notifications were only shown once, enabling users to access their notifications at any time and significantly improving system reliability and engagement.

Mine Alliance Website | *Hackathon Project - 100+ Participants* | ∼50 hours | github.com/LuaanNguyen/Mine-Alliance

- Designed a **Flask backend forum application** with user authentication and integrated AI chatbot support using the **OpenAI API** for summarization, assessment, and information delivery to assist users and stakeholders with actionable mining queries.
- Provided an end-to-end solution empowering individuals, companies, and government authorities to collaborate and resolve mining issues more effectively, supporting informed, transparent decision-making and achieving a **$1.5K hackathon prize**.

Sage AI - Data Science Tutor | *Hackathon Project* | ∼40 hours | github.com/sbeeredd04/sage

- Developed a multi-user **AI tutoring platform** with **React, Flask, Docker**, and GPU acceleration, integrating **OpenAI, Gemini, and Claude APIs** for real-time web-grounded results and leveraging a **RAG pipeline** for accuracy.
- Implemented GPU cluster deployment and benchmarking tools for collaborative code execution, and built an immersive, sandboxed learning environment using the **Monaco Editor**.

Activities

Director of Technology | Software Developers Association (SoDA) | Jan 2024 – Present | thesoda.io

- Built and scaled SoDA's internal tools platform (including a **Discord OAuth2.0-secured admin site**, summarizer bot, and member rewards/storefront) to support **3,000+ members** across multiple clubs (ACM, WiCS), enabling secure and automated management.
- Led an **11-member team** to deliver **12+ technical workshops** and hackathons (Docker, Chrome Extensions, Flask), removing roadblocks and fostering a creative, collaborative culture, empowering **600+ active students** with hands-on software engineering skills.

Hackathon Achievements | *Various Hackathons* | 2022 – Present

- **Won 5 hackathons**, earning over **$8K** in prizes and leading cross-functional team projects and MVPs under pressure; demonstrated strong teamwork, adaptability, and problem-solving skills.

Skills

Languages: Python, C/C++, Java, JavaScript, TypeScript, C#, SQL
Frameworks & Libraries: Flask, FastAPI, Django, React, Next.js, Node.js, Express, PyTorch, TensorFlow, LangChain, LangGraph, Three.js
AI/ML & Data: Computer Vision, VLM models, Gemini API, OpenAI API, Claude API, Hugging Face, Wandb, CUDA
Cloud, DevOps & Tools: AWS, Docker, Vercel, Linux, Unix, Git, GitHub, CI/CD, Jupyter, XCode, Redis, Celery, REST APIs